

Deep Learning — Week 7 Notes

Bias–Variance Tradeoff · L2 Regularization · Early Stopping · Dataset Augmentation · Dropout · Ensembles

NPTEL / IIT Ropar · CS7015 Lecture 8 · Exam-focused revision sheet

★ Includes full worked answer key (all 20)

1 · Bias–Variance Tradeoff

Decomposing the expected error

$$E[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2 \text{ (irreducible noise)}$$

- **Simple model** → high bias, low variance (underfits).
- **Complex model** → low bias, high variance (overfits; predictions swing a lot across training sets).
- Large variation in predictions across different training subsets = **high variance**. The cure is **regularization**.

Case Study 1 — simple linear model vs flexible NN, true demand = 48; NN predictions across 4 subsets average ≈ 47 : simple linear model = **high bias, low variance** · average NN prediction ≈ 47 · squared bias = $(48-47)^2 = 1$ · large NN variation $\Rightarrow$ **high variance under the case assumptions** · best fix $\Rightarrow$ **add a suitable regularization method**.

Note on CS1 numbers: the source screenshot was low-resolution; the average-prediction / squared-bias pair is computed as $(\text{true} - \text{mean prediction})^2$. If the four predictions average to 46 instead, squared bias = 4. The method is what matters — verify the four printed predictions on your copy.

2 · L2 Regularization (Weight Decay)

Penalize large weights

$$\tilde{L}(w) = L(w) + (\alpha/2) \sum w_i^2 \rightarrow \partial \tilde{L} / \partial w = \partial L / \partial w + \alpha w$$

- Adds a penalty proportional to weight magnitude $\rightarrow$ **discourages large weights**, shrinking them toward zero.
- The regularization contribution to the gradient is $\alpha \cdot w$; the full gradient is data-gradient + αw .
- Effect: reduces variance / overfitting $\rightarrow$ **validation performance may improve**.

Case Study 2 — $w = 5$, data-loss gradient = 4, $\alpha = 2$, $\eta = 0.1$: purpose of L2 = **discourage large weights** · regularization contribution = $\alpha w = 2 \times 5 = 10$ · full regularized gradient = $4 + 10 = 14$ · updated $w = 5 - 0.1 \times 14 = 3.6$ · expected outcome = **validation performance may improve due to reduced overfitting**.

3 · Early Stopping

Stop when validation error starts rising

Track **validation** loss during training. Training loss keeps falling, but validation loss falls then **rises** once the model starts overfitting. Stop at the validation minimum — this effectively limits model complexity (a form of regularization).

Epoch	Training loss	Validation loss
5	0.62	0.5x
10	0.44	0.3x
15	0.31	0.13 (min)
20	0.22	rising
25	0.16	0.51 (high)

Case Study 3 — energy-consumption model: validation minimum at **Epoch 15** · overfitting after epoch 15 shows as **training loss decreasing while validation loss increases** · monitor validation because it **reflects performance on unseen data** · early stopping **acts as a regularization method by limiting over-training** · keeping epoch 25 over 15 risks **the model generalizing worse despite lower training loss**.

4 · Dataset Augmentation, Dropout & Ensembles

More effective data + averaging away variance

- **Augmentation:** generate extra training examples via label-preserving transforms (rotations, shifts, brightness, zoom) → exposes the model to realistic input variations.
- **Dropout:** randomly keep each unit with probability p during training → on average $p \cdot (\text{layer size})$ units stay active; acts like training an ensemble of thinned networks.
- **Ensembles:** average several independently trained models → **reduces variance** by combining diverse predictions.

Case Study 4 — 900 images, 70/30 train/val split, 2 augmented images per original, dropout keep $p=0.8$ on a 50-unit layer, 3 CNNs averaged:

original training images = $900 \times 0.7 = 630$ · effective size after augmentation = $630 \times 3 = 1890$ · augmentation helps because it **exposes the model to realistic input variations** · active units under dropout = $50 \times 0.8 = 40$ · averaging 3 CNNs helps by **reducing variance through combining diverse predictions**.

★ FULL ANSWER KEY — Week 7 (assignment not submitted)

Q	Answer	Q	Answer
1	High bias, low variance	11	Epoch 15
2	≈ 47 (mean of predictions)*	12	Training loss ↓ while validation loss ↑
3	1 (squared bias)*	13	Validation reflects unseen-data performance
4	High variance under the assumptions	14	Regularization by limiting over-training
5	Add a suitable regularization method	15	May generalize worse despite lower train loss
6	Discourage large weights	16	630
7	10 ($= \alpha w$)	17	1890
8	14 (full gradient)	18	Exposes model to realistic input variations
9	3.6 (updated w)	19	40 ($= 0.8 \times 50$)
10	Validation may improve (less overfit)	20	Reduces variance via diverse predictions

* Q2/Q3 depend on the four printed predictions, which were low-res in the source. Formula: mean of predictions, then squared bias = $(48 - \text{mean})^2$. Double-check the four numbers on your copy.